
Agent RAGLAW: Bottleneck-Typed Retrieval for Legal and Multi-Hop Question Answering

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Abstract

Retrieval-augmented generation (RAG) is often presented as a broadly helpful way to ground language models, but our experiments on legal and multi-hop question answering show a more conditional picture. We began with a full LangGraph agent for legal RAG, using planning, retrieval, synthesis, evaluation, and replanning. Early experiments showed that this heavy agentic pipeline could underperform direct prompting, especially when the model already had strong legal priors. We therefore reframed the project around a simpler question: when is the bottleneck retrieval, and when is it reasoning or answer selection? We introduce a bottleneck-typed evaluation of RAG methods across BarExam, MuSiQue, CaseHOLD, LegalBench-SCALR, and HousingQA. A simple retrieval-depth diagnostic separates regimes: MuSiQue with Llama 70B collapses when retrieval is restricted from top-5 to top-1 passages (27.5% to 13.0%, -14.5pp, McNemar $p = 4.18 \times 10^{-7}$), while BarExam with Gemma 4 26B is flat under the same ablation (82.5% to 83.0%, $p = 1.00$). We use snap-conditioned retrieval as the main method vehicle: the model first commits to a tentative answer and a HyDE-style passage, retrieves on that passage, and then synthesizes a final answer. On MuSiQue, a two-call snap-HyDE variant improves Llama 70B from 27.5% to 37.0% (+9.5pp, $p = 0.0079$). On BarExam, snap+HyDE improves Gemma 4 26B from 78.08% to 81.17% at full corpus ($N = 1195$), with the smaller Gemma 4 E4B showing the same direction. The central lesson is not that one RAG method always wins, but that retrieval methods should be selected according to the benchmark-model bottleneck.

1 Introduction

Legal question answering is a difficult testbed for RAG. The model must apply rules, exceptions, and fact patterns precisely, and the retrieved text may be only partially aligned with the issue in the question. Our original project therefore built an agentic legal RAG system: a classifier/planner decomposed a question, an executor retrieved and synthesized evidence for each step, an evaluator judged whether evidence was sufficient, and a replanner added or revised steps when the evidence was incomplete.

The first lesson was negative but useful. The full agentic loop was expensive and did not reliably outperform direct model answering. On the midterm BarExam sample, the agentic pipeline underperformed direct prompting for both a mid-size Gemma model and DeepSeek Reasoner. This failure was not simply an implementation mistake; it exposed a broader problem. More retrieval and more

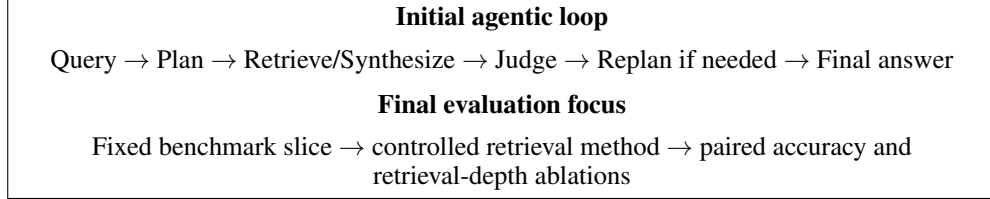


Figure 1: Project pivot. The initial architecture emphasized a full legal research agent. The final experiments emphasize controlled method comparisons and bottleneck diagnostics.

agent structure can distract a model that already has strong priors, while also fragmenting reasoning across multiple calls.

We pivoted from “build the largest agent” to “measure which bottleneck is active.” The result is a bottleneck-typed account of RAG. Some benchmark-model pairs are *retrieval-bottlenecked*: answer accuracy is sensitive to whether enough useful passages are retrieved. Others are *reasoning-bottlenecked*: the model’s legal reasoning or answer calibration dominates, and additional passages have little effect or can even harm. A third regime appears in multiple-choice holding selection tasks, where retrieval may pull competing candidate holdings rather than clarifying the decision.

This framing also explains why a seemingly weak control, `golden_passage`, can underperform `llm_only` and `rag_snap_hyde` on BarExam. Providing a single labeled passage is not an oracle ceiling. In our full Gemma 4 26B BarExam audit, `golden_passage` flips 96 previously correct `llm_only` answers to wrong and only 83 wrong answers to correct. The gold passage often anchors the model on a related but incomplete rule. In contrast, snap-first methods let the model commit to a legal prior before reconciling with evidence, which is why `snap_only_in_final` and `rag_snap_hyde` both beat the forced gold-passage control.

2 Related Work

Our method is related to generated-query and draft-conditioned retrieval. HyDE generates a hypothetical document and embeds it for zero-shot dense retrieval [1]. Query2Doc similarly uses LLM-generated pseudo documents for query expansion [2]. FLARE retrieves using tentative generated text during long-form generation [3], while Iter-RetGen and CoRAG iterate between generation and retrieval [4, 5]. Speculative RAG uses multiple retrieved document subsets to draft answers and a verifier to select among them [6]. Our snap-HyDE variant is simpler and fixed-cost: one call produces a tentative answer and a retrieval passage, and a second call synthesizes the final response.

The more important distinction is the bottleneck framing. Adaptive-RAG routes questions by complexity [7]; Know Your RAG and GRADE classify question difficulty and multi-hop structure [8, 9]. Legal RAG benchmarks have already shown that retrieval helps some legal tasks and hurts others. Zheng et al. introduce BarExam QA and HousingQA and report that retrieval helps Housing much more than BarExam [10]. Vaddi reports task-dependent legal RAG behavior, including gains on ContractNLI and degradation on CaseHOLD [11]. We build on these observations by treating retrieval benefit as a measurable bottleneck signature rather than a universal property of the retriever.

3 System and Method

3.1 Initial agentic legal RAG system

The initial system was a LangGraph plan-execute-evaluate loop. A planner constructed a legal research plan, an executor ran multi-query retrieval and source-grounded synthesis for each step, a judge checked whether the retrieved evidence was sufficient, and a replanner added follow-up research steps when needed. This design was useful for exploring legal-agent architecture, but the evaluation showed that its many moving pieces made it hard to identify where accuracy gains or losses came from.

Table 1: BarExam Tier 3 method matrix: Gemma 4 26B-A4B, $N = 1195$.

Method	Accuracy	Δ vs rag_simple
rag_simple	78.08%	–
golden_passage	78.66%	+0.58pp
rag_hyde	78.91%	+0.83pp
llm_only	79.75%	+1.67pp
snap_only_in_final	80.59%	+2.51pp
rag_snap_hyde	81.17%	+3.09pp

3.2 Snap-conditioned retrieval

The core method family is snap-conditioned retrieval. In the legal multiple choice setting, the model first produces a tentative answer or legal analysis. That snap is then used to guide a HyDE-style passage or evidence query. The retriever searches with this answer-shaped query, and a final synthesis call combines the original question, snap, and retrieved passages.

The two-call version, `rag_hyde_2call`, makes this fixed-cost:

1. Call 1: produce a tentative answer plus a reference passage stating the controlling rule, doctrine, fact, or principle.
2. Retrieval: embed the generated passage and retrieve the top- k passages from the corpus.
3. Call 2: synthesize the final answer using the question, snap, and retrieved evidence.

This design is not claimed as a universal invention independent of prior work. Its value in this project is as a compact probe: when it helps, we can ask whether the gain came from better retrieval recall, stronger answer anchoring, or both.

3.3 Bottleneck diagnostic

We use retrieval-depth sensitivity as a simple diagnostic. If reducing `rag_simple` from top-5 to top-1 retrieved passages sharply reduces accuracy, the benchmark-model pair is retrieval-bottlenecked. If the result is flat, additional retrieval depth is not the active bottleneck. This diagnostic is method-independent because it can be applied before testing more elaborate retrieval shaping.

4 Experimental Setup

Datasets. We evaluate on BarExam QA, MuSiQue, CaseHOLD, LegalBench-SCALR, MLEB-SCALR, and HousingQA. BarExam is a legal multiple-choice benchmark over bar-exam fact patterns. MuSiQue is open-ended multi-hop QA. CaseHOLD and SCALR test holding selection among candidate legal holdings. HousingQA tests yes/no statutory reasoning over state housing law. MLEB-SCALR is used as a retrieval-only calibration benchmark with qrels.

Models. The main legal full-corpus results use Gemma 4 26B-A4B and Gemma 4 E4B served through the cluster evaluation path. The main multi-hop results use Llama 3.3 70B. We also use Gemma 3 27B as an important negative cross-family check on MuSiQue.

Evaluation. For answer accuracy, we report exact match or multiple-choice accuracy depending on the dataset. For paired comparisons, we use McNemar tests on the same question slice. For retrieval-only SCALR, we report Recall@k, MRR@k, and nDCG@k. We treat small samples as directional and emphasize full-corpus or paired $N = 200$ results when drawing conclusions.

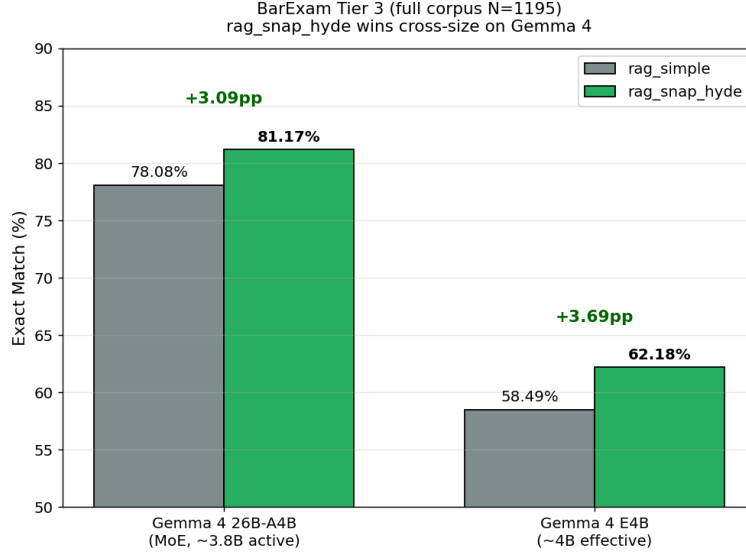


Figure 2: BarExam full-corpus cross-size result. rag_snap_hyde improves over rag_simple on both Gemma 4 model sizes.

Table 2: Llama 70B MuSiQue paired results, $N = 200$.

Method	EM	Δ vs rag_simple	McNemar p
rag_simple	27.5%	–	–
rag_multi_query	29.0%	+1.5pp	0.728
multi_hyde_diverse	35.5%	+8.0pp	0.0195
iterative_planning_table	36.0%	+8.5pp	0.0533
snap_hyde_2call	37.0%	+9.5pp	0.0079
subagent_rag	15.5%	-12.0pp	0.0007

5 Results

5.1 BarExam: snap-first reasoning beats raw retrieval and gold context

The BarExam result is the cleanest full-corpus legal finding. On Gemma 4 26B-A4B, rag_snap_hyde reaches 81.17%, beating rag_simple by +3.09pp. The same winner appears on Gemma 4 E4B, improving from 58.49% to 62.18% (+3.69pp). Figure 2 shows the cross-size pattern.

The important mechanism is that BarExam is not simply “retrieval helps.” In fact, llm_only beats golden_passage, and snap_only_in_final is close to the best method. Our audit showed that gold-passage injection is a noisy single-snippet control: it helps some cases but also anchors the model to partial or misleading doctrine. The snap-first methods work because they preserve the model’s strong legal prior before asking it to reconcile with retrieved text.

5.2 MuSiQue: retrieval depth is first-order

MuSiQue behaves differently. The Llama 70B baseline rag_simple achieves 27.5% EM. snap_hyde_2call reaches 37.0%, a +9.5pp improvement with paired McNemar $p = 0.0079$. multi_hyde_diverse and iterative_planning_table also improve substantially, while the current subagent_rag implementation is significantly worse because its gap-routing prompts over-abstain.

The retrieval-depth ablation shows why this is a retrieval-bottlenecked regime. Restricting rag_simple from top-5 to top-1 passages drops EM from 27.5% to 13.0% (-14.5pp, $p = 4.18 \times 10^{-7}$). Across MuSiQue methods, top-1 causes large drops of roughly 10–17pp. Passage count is

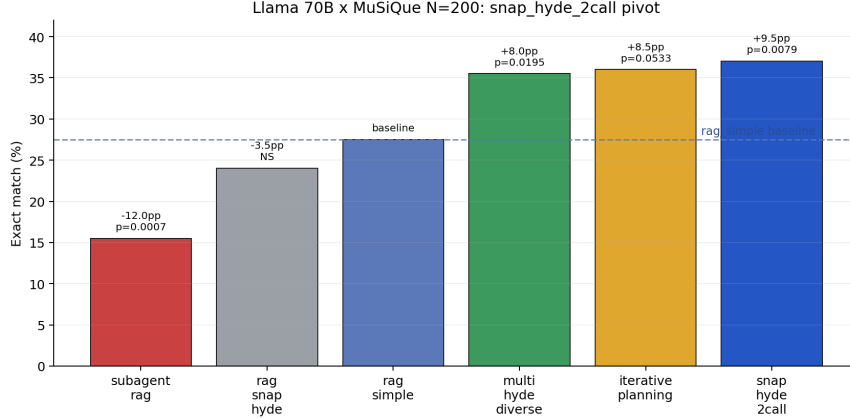


Figure 3: Current MuSiQue pivot for Llama 70B, $N = 200$. The two-call snap-HyDE variant is the strongest significant method in this slice; the current subagent routing implementation over-abstains.

Table 3: Retrieval-depth and method behavior across datasets.

Dataset / model	Key diagnostic	Observed behavior	Bottleneck read
MuSiQue / Llama 70B	top-5 to top-1 <code>rag_simple</code>	27.5% to 13.0%	Retrieval-bottlenecked: multi-hop needs multiple passages.
BarExam / Gemma 4 26B	top-5 to top-1 <code>rag_simple</code>	82.5% to 83.0%	Reasoning/anchoring-bottlenecked: retrieval depth is flat.
CaseHOLD / Llama 70B	top-5 to top-1 and 2-call	72.0% to 70.5%; 2-call 69.5%	Option disambiguation: retrieval can surface competing holdings.
LegalBench-SCALR / Llama 70B	top-5 to top-1	77.0% to 59.5%	Candidate-depth limited: a small candidate set helps, but top-10 adds little over top-5.
HousingQA / Gemma 4 26B	top-k sweep and state filtering	in progress	Predicted statutory retrieval bottleneck; state metadata is now wired.

therefore a first-order variable; method choice matters, but only after the system retrieves enough evidence.

5.3 Cross-dataset bottleneck taxonomy

This table is the main conceptual result. Legal QA is not one regime. BarExam is flat under retrieval depth, while SCALR is sharply candidate-depth limited. CaseHOLD appears answer-flat under current runs, though we repaired the gold-option retrieval mapping so future runs can measure gold-hit properly. HousingQA is the next statutory test: the harness now has state-filtered retrieval so we can ask whether matching the question’s state is enough to recover accuracy.

6 Mechanism Analysis

6.1 Why the gold passage can be worse than snap-HyDE

The BarExam golden-passage result initially looks paradoxical: if the system is given the gold passage, why does it underperform `llm_only` and `rag_snap_hyde`? The answer is that the gold passage is a control, not an oracle. A single passage can be topically related without containing the decisive rule, or it can emphasize a distractor doctrine that is plausible but wrong for the fact pattern.

In the paired Gemma 4 26B BarExam audit, `llm_only` gets 79.75% while `golden_passage` gets 78.66%. The transition table shows 96 `llm_only`-correct answers flipped wrong by the gold passage, compared with 83 `llm_only`-wrong answers fixed by the passage. Meanwhile, `rag_snap_hyde` reaches 81.17%. The mechanism is snap anchoring: the model commits to a legal

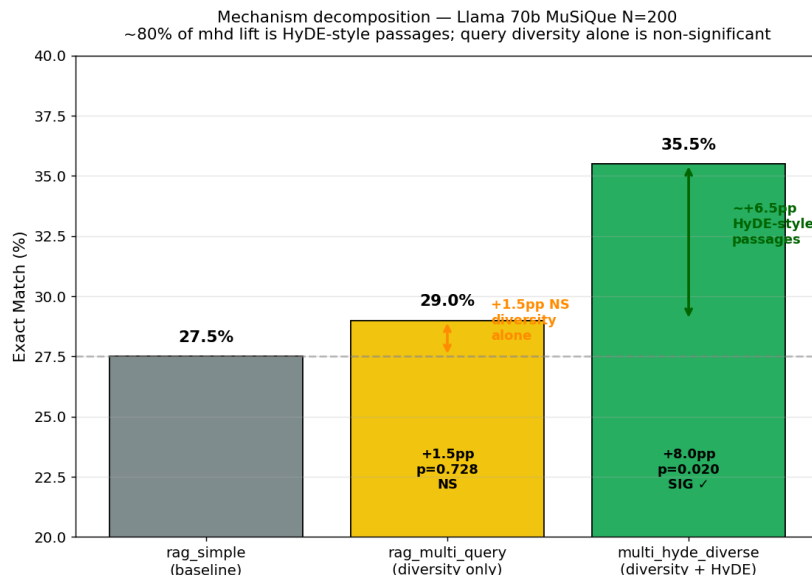


Figure 4: MuSiQue mechanism decomposition. Query diversity alone is small; the larger gain comes from answer-bearing HyDE-style passages.

answer before retrieval can over-anchor it on a single snippet. The retrieval component still adds marginal value, but most of the BarExam lift comes from snap-first reasoning.

6.2 Why snap-HyDE is model-conditional on MuSiQue

The MuSiQue story is more retrieval-dominant. Llama 70B’s snap-HyDE passages improve gold-passage retrieval, and EM rises. Gemma 27B’s snap-HyDE passages move retrieval away from gold passages, and EM falls. This explains why `snap_hyde_2call` is not a model-agnostic multi-hop cure: the method helps only when the model can write a generated passage that matches the corpus style and points retrieval toward the right evidence.

6.3 Cost and practicality

The final methods are intentionally much cheaper than the first agentic loop. The full LangGraph agent used many calls per question and made debugging difficult. `snap_hyde_2call` fixes the budget at two LLM calls per question, while `multi_hyde_diverse` and `planning-table` methods spend more calls. Figure 5 shows that the useful methods are not just the highest-accuracy methods; cost per question matters.

7 Limitations

The strongest MuSiQue result is paired $N = 200$, not a full-corpus run. It is statistically significant, but a larger replicate would make the claim more robust. The BarExam result is full-corpus and cross-size, but its effect size is modest; it should be presented as a mechanism result rather than a dramatic accuracy breakthrough. CaseHOLD previously lacked meaningful gold-option retrieval instrumentation; we repaired the mapping, but new retrieval claims require a rebuilt Chroma collection and fresh logs. HousingQA state-filtered retrieval has been wired and submitted, but its result is not included here.

The method is also provider- and model-dependent. OpenRouter-served Gemma runs showed runaway-loop behavior on iterative prompts, so Gemma results should prefer cluster-served vLLM logs. More broadly, snap-HyDE depends on the model’s ability to generate a useful intermediate passage. If the generated passage is off-style or wrong, retrieval can get worse.

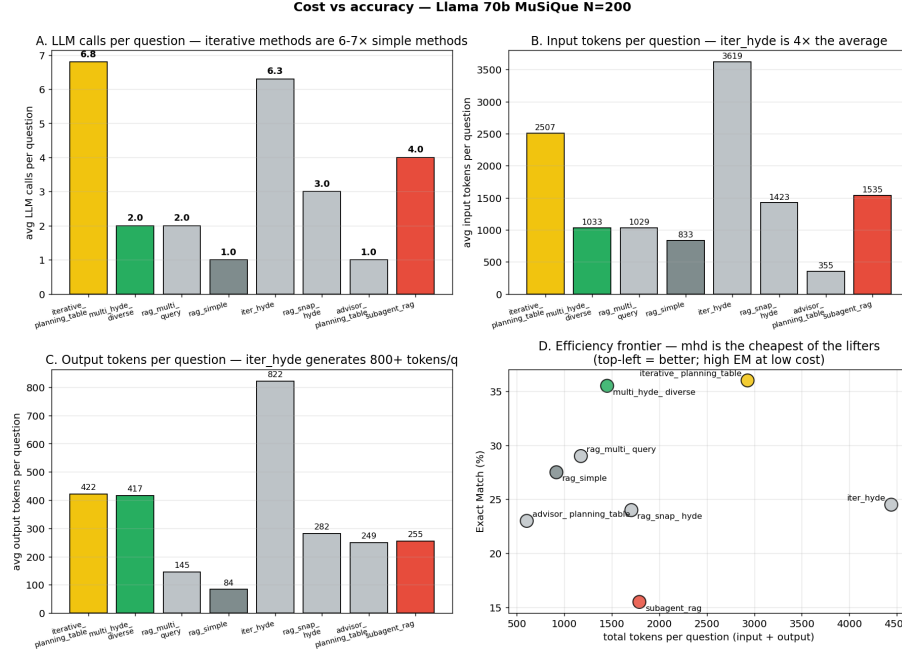


Figure 5: Cost vs. accuracy on Llama 70B MuSiQue. Fixed-cost snap-HyDE and multi-HyDE methods are more practical than high-call iterative agents.

8 Conclusion

The final project result is a more nuanced account of agentic legal RAG than the original proposal. Heavy planning agents are not automatically better than simple baselines. Retrieval is not automatically helpful, and gold context is not an oracle. The useful direction is bottleneck-aware method selection: multi-hop QA needs enough retrieved evidence and benefits from answer-shaped retrieval; BarExam-style legal MC benefits more from snap-first reasoning and careful reconciliation; holding-selection tasks need candidate-set diagnostics; statutory QA likely needs metadata-aware retrieval. This is the strongest story for a class report and the best foundation for future work on adaptive legal RAG systems.

References

- [1] Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. Precise zero-shot dense retrieval without relevance labels. *ACL*, 2023.
- [2] Liang Wang, Nan Yang, and Furu Wei. Query2Doc: Query expansion with large language models. *EMNLP*, 2023.
- [3] Zhengbao Jiang et al. Active retrieval augmented generation. *EMNLP*, 2023.
- [4] Zhihong Shao et al. Enhancing retrieval-augmented large language models with iterative retrieval-generation synergy. *EMNLP Findings*, 2023.
- [5] Zhiruo Wang et al. CoRAG: A chain-of-retrieval augmented generation framework. *arXiv:2501.14342*, 2025.
- [6] Zilong Wang et al. Speculative RAG: Enhancing retrieval augmented generation through drafting. *ICLR*, 2025.
- [7] Soyeong Jeong et al. Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity. *NAACL*, 2024.
- [8] IBM Research. Know Your RAG: Dataset and taxonomy for retrieval augmented generation. *COLING*, 2025.
- [9] GRADE. Generating multi-hop QA with a reasoning-depth and semantic-distance difficulty matrix. *EMNLP Findings*, 2025.

- [10] Lucia Zheng, Neel Guha, and collaborators. A reasoning-focused legal retrieval benchmark. *CS&Law / arXiv:2505.03970*, 2025.
- [11] Vaddi. Can small models reason about legal documents? *arXiv:2603.25944*, 2026.
- [12] Harsh Trivedi et al. MuSiQue: Multihop questions via single-hop question composition. *TACL*, 2022.
- [13] Neel Guha et al. LegalBench: A collaboratively built benchmark for measuring legal reasoning in large language models. *arXiv:2308.11462*, 2023.